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Final Scheme/ Answer Key for Valuation

Scheme of evaluation (marks in brackets) and answers of problems/key

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

FIRST SEMESTER B.TECH DEGREE EXAMINATION, DECEMBER 2019

Course Code: CYT100

Course Name: ENGINEERING CHEMISTRY

(2019-Scheme)

Max. Marks: 100 Duration: 3 Hours

PART A

Answer all questions, each carries 3 marks. (3)

1 Nernst equation,substitution-2 marks,answer-1 mark (3)

2 Reduction: $H^+ + e \rightarrow 1/2H_2$;E^0 =0V (3)

Oxidation :$Fe \rightarrow Fe^{2+} + 2e$; E^0=-0.44V

E^0_{cell} =0--0.44 V =0.44V

For $Fe \rightarrow Fe^{3+} + 3e$; E^0=-0.04V

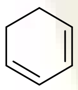
E^0_{cell} =0--0.04 V =0.04V

Ferrous ion formation is more spontaneous than ferric as cell emf is greater(ΔG=-nFE) -1 mark, Equation-2 marks. $Fe + 2HCl \rightarrow FeCl_2 + H_2$or redox equations separately.

OR consider H_2 evolution (1 mark)Oxidation of Fe (1 mark)

3 Beer-Lamberts law, Statement-2 marks, any one limitation-1 mark. (3)

4 (3)

a)  conjugation(2 marks), will absorb at higher wavelength (1 mark).

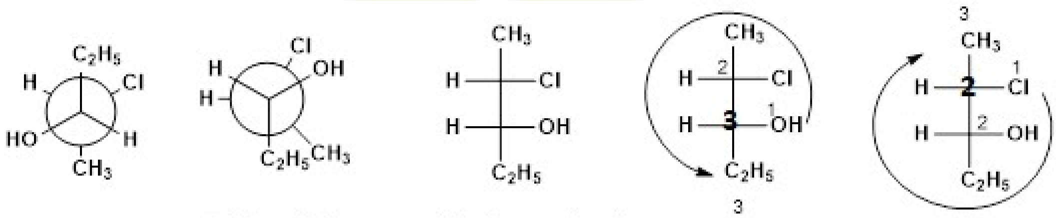
5 Mobile phase liquid or gas, Stationary phase solid or liquid OR (3)

S-L,S-G,L-L,L-G

Adsorption, partition and ion-exchange can be given 2 marks

6 Chemical reduction using reducing agent (2 marks) example 1 mark (3)

7 Configuration of C2 – **1.5** mark, C3 – **1.5** mark, (3)



Eclipsed form Fischer projection 2s, 3R

2(S)-chloropentan-3(R)-ol. Incorrect IUPAC name do not reduce marks. Correct

IUPAC name is given one mark though notations are wrong.

8 Random, alternating, block and graft with AB. Any 3 -3 marks. (3)

9 i) Normality=0.05 x 3 =0.15N (3)

Hardness= 0.15 x 50 =7.5g/L= 7500ppm (1 Marks)

Hardness= .04 x 50= 2g/L=2000ppm (2 marks)

Any one correct answer 2 marks

10 Any 2 significance-3 marks (BOD definition 1 mark) (3)

PART B

Answer one full question from each module, each question carries 14 marks

Module-I

11 a) Construction-2 marks,electrode reaction-2 marks,Nernst eqn-2 marks,E value at different KCl concentrations-2 marks (As chloride concentration increases, E value decreases may be given marks) (8)

b) $H^+ + e \rightarrow \frac{1}{2}H_2; E^0 = 0$ (When pH=0); $E = -0.0591$ pH; (6)

$H^+ + e \rightarrow \frac{1}{2}H_2; E = -0.83$ (When pH=14);

Iron electrode has higher reduction potential than hydrogen electrode in alkaline medium. Hence iron will not oxidise in alkaline medium. Mg will react with both acids and alkali as its reduction potential is lower than hydrogen electrode potential in both medium

Acid medium (2marks) , Nernst's equation for hydrogen electrode 2 marks. alkaline medium (2 marks)

12 a) Construction-2 diagram 1 mark, charging or discharging reactions-4 marks advantages 1 mark (8)

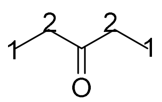
b) At cathode (-ve) ions approaching are Na^+ and H^+. As H^+ has higher reduction potentials. It undergoes reduction to H_2 gas.-(3) (6)

At anode(+ve) ions approaching are Cl^- and OH^-. But Cu electrode is also there.So Cu with lower reduction potential undergoes oxidation to Cu^{2+} -(3)

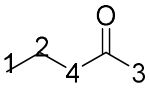
Liberation of Cl_2 or O_2 may be given 1 mark.

Module-II

13 a) 4+4=8

(i) 

two signals, CH_3 protons and CH_2protons,relativepositions,Splitting pattern of CH_3(triplet)and CH_2 (quartet).

and (ii)  four signals, relative positions,splitting pattern 1 (triplet), 2(hextet or sextet), 3 (singlet), and 4(triplet)

b) $k = 4\pi^2 c^2 \overline{\nu}^2 \mu$ (6)

$$\frac{k_1}{k_2} = \frac{\overline{\nu}_1{}^2 \mu_1}{\overline{\nu}_2{}^2 \mu_2} = \frac{3000^2 \times \frac{1\times12}{1+12}}{1700^2 \times \frac{16\times12}{16+12}} = 0.419$$

(Equation-3,sub-2, Answer-1)

Separate force constant calculation can also be given marks.

14 a) Instrumentation-3 components-3 , Increasing wavelength of absorption with conjugation with suitable examples-2 marks (8)

b) Principle-4 marks,applications-2 marks (6)

Module-III

15 a) Instrumentation with labelling-4 marks each. (8)

b) Name of two detectors each for GC and HPLC -3 marks each (6)

GC –Thermal conductivity dictator, flame ionization detector, electron capture detector etc.

HPLC- Bulk property detector (eg refractive index) , solute property detector(UV dictator)

16 a) Principle-3marks, instrumentation-3 marks,appln-2 marks (8)

b) 3 points of difference-2marks for each (6)

Module-IV

17 a) Ethane-4marks-[Conformers-1,energy diagram-2,more stable-1] Butane-6 marks[Conformers-2,energy diagram-3more stable-1] (10)

b) Classification-4marks (4)

18 a) Definition-2 marks, Cis,trans-2 marks,conformations-3 marks, stability-1 mark (8)

b) Principle-2 marks, construction-2 marks, working- 2 marks (6)

Module-V

19 a) Primary-3 marks, secondary(trickling filter and UASB)-5 marks, tertiary-2 marks (10)

b) Any three disadvantages - 4marks (4)

20 a) Principle-3 marks, procedure-5 marks (8)

b) COD = [(V_1-V_2) x N x 8/ Ve] x 1000 (6)

V_1-Vol. of ferrous solution for blank, V_2 - Vol. of ferrous solution for sample, N – Normality of ferrous solution,

V_e – Volume of sample used

COD = [(20.4 – 12.4) x 0.2 x 8/50] x 1000 = 256 mg/L

Equation (2 mark), substitution (2 mark), answer (2 marks)

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